Google Colab Overview

Google Colaboratory, or Google Colab in short, is a cloud-based platform that allows users to write and execute Python code in a web-based Jupyter notebook environment. It offers free access to powerful computing resources, including GPUs and TPUs, making it ideal for machine learning and data analysis tasks. Colab also integrates with Google Drive for easy file management and sharing.

There are multiple advantages to using Google Colab over other programming IDEs (integrated development environments).

- **Absence of setup requirements**: Installing IDEs on a local system can often be a hassle due to system-specific issues, which may be difficult to resolve. Google Colab, being browser-based, eliminates these setup challenges.
- **Maintenance of libraries/packages**: Managing library/package versions to resolve dependencies can be complex and system-specific. Google Colab alleviates this as the libraries/packages are maintained at compatible versions by Google, reducing dependency-related issues.
- **Execution time:** Google Colab offers higher CPU computing power than most local systems and free (but limited) access to GPUs and TPUs. This enables faster code execution, which is particularly beneficial for tasks such as hyperparameter tuning in machine learning models.

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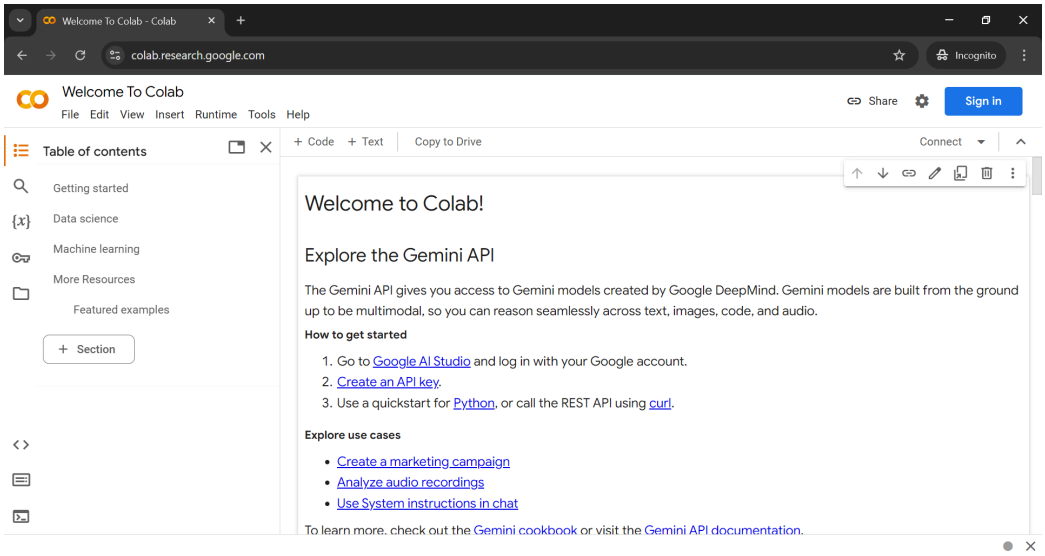
Accessing Google Colab

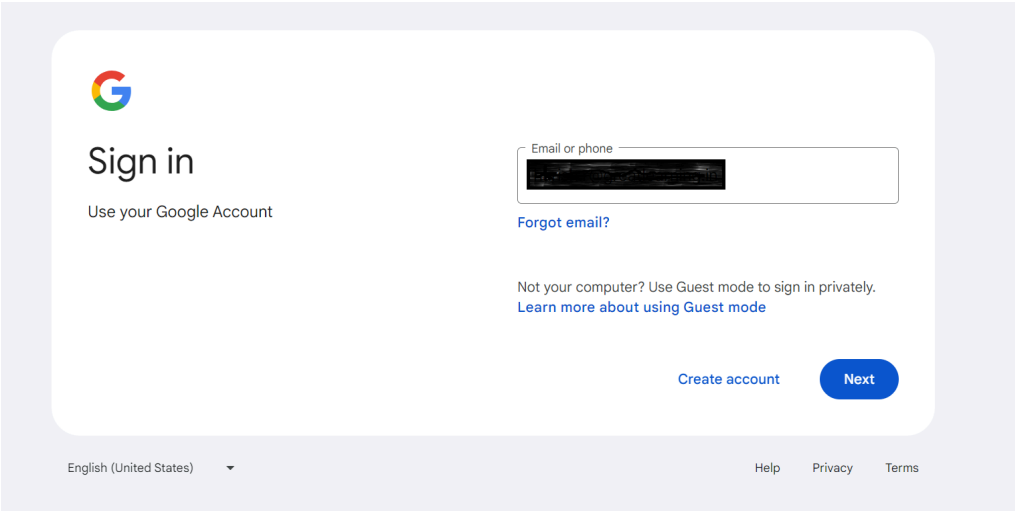
1. Open this link in a web browser. It should open up the following window.



2. If you're not already signed into a Google account, click on the **"Sign in"** button in the top right corner and log in with your Google account.

Note: If you're already signed in to your Google account, you are likely to be automatically logged in to Google Colab.

3. In case you are not signed into your Google account, enter your credentials and click ***Next*** to sign in.



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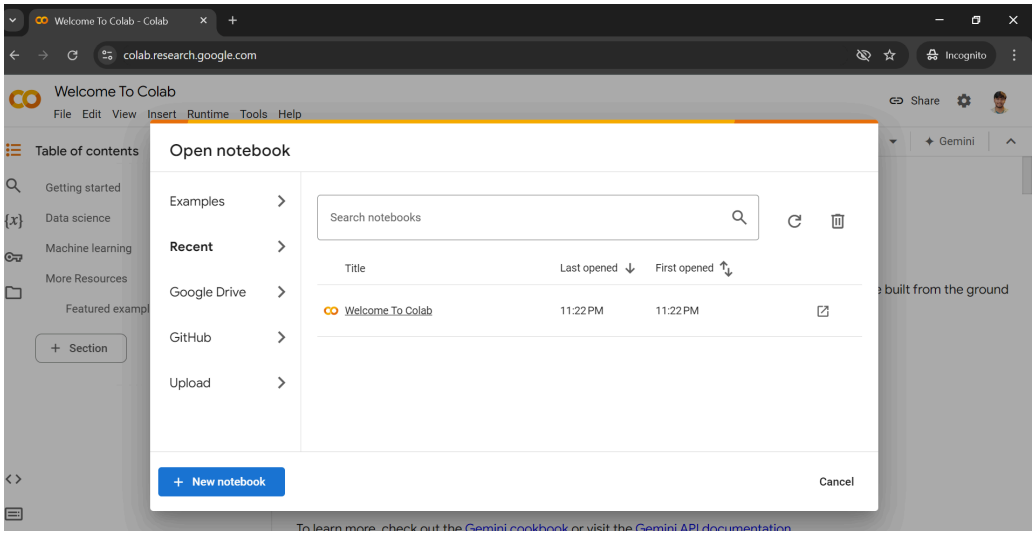
Note: If you have 2-step verification enabled on your Google account, after entering your password, you will be prompted to complete the additional verification step. Follow the instructions to complete the verification. Once you have completed the verification process, you will be successfully logged into Google Colab.

4. After logging in, you will be on the Google Colab homepage.



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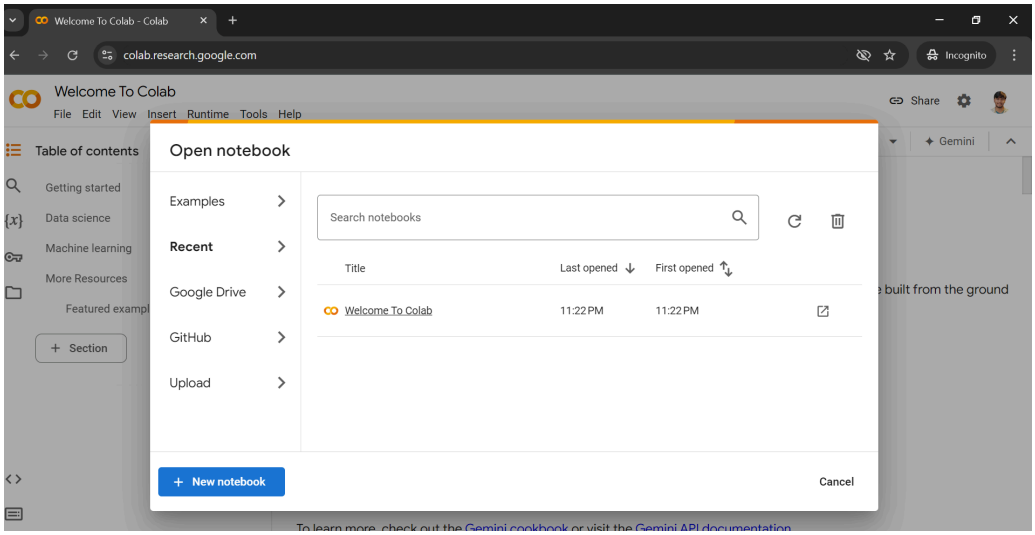
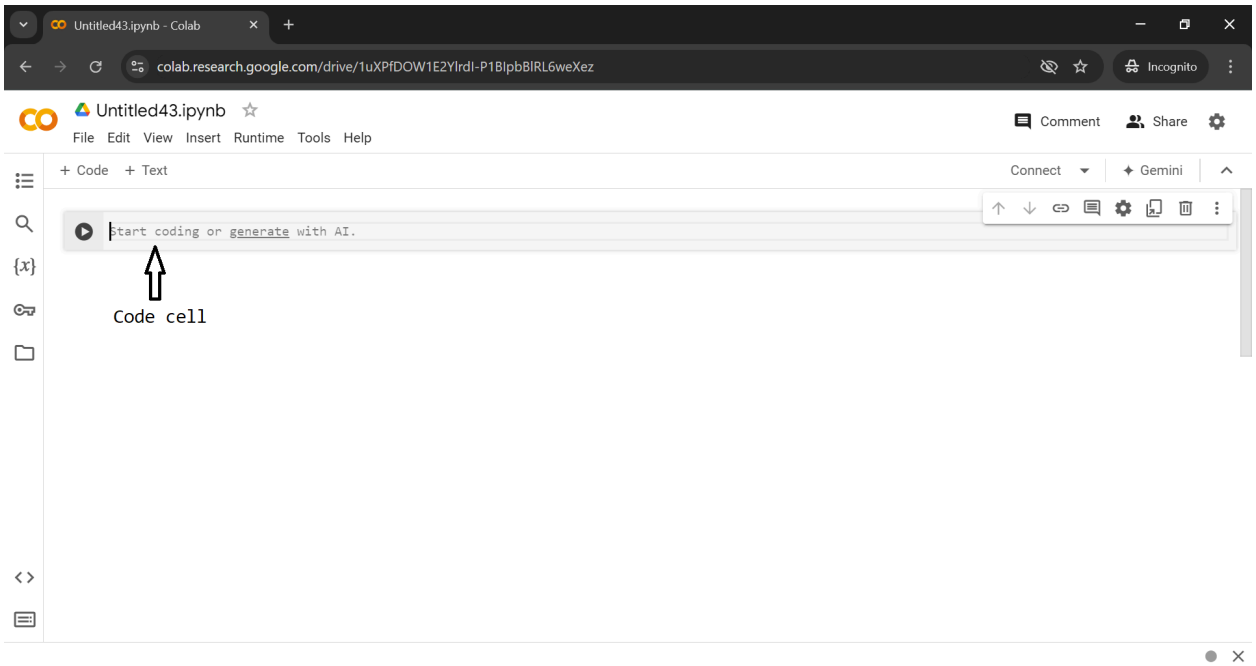
Creating a new Python notebook

1. Click on **'+ New Notebook'** to create a new Colab notebook.



Note: Kindly be patient while Google Colab configures everything; it may take a few seconds to create a new notebook for you.

2. This will open up a new Python notebook. You can now start writing your code in the code cell.



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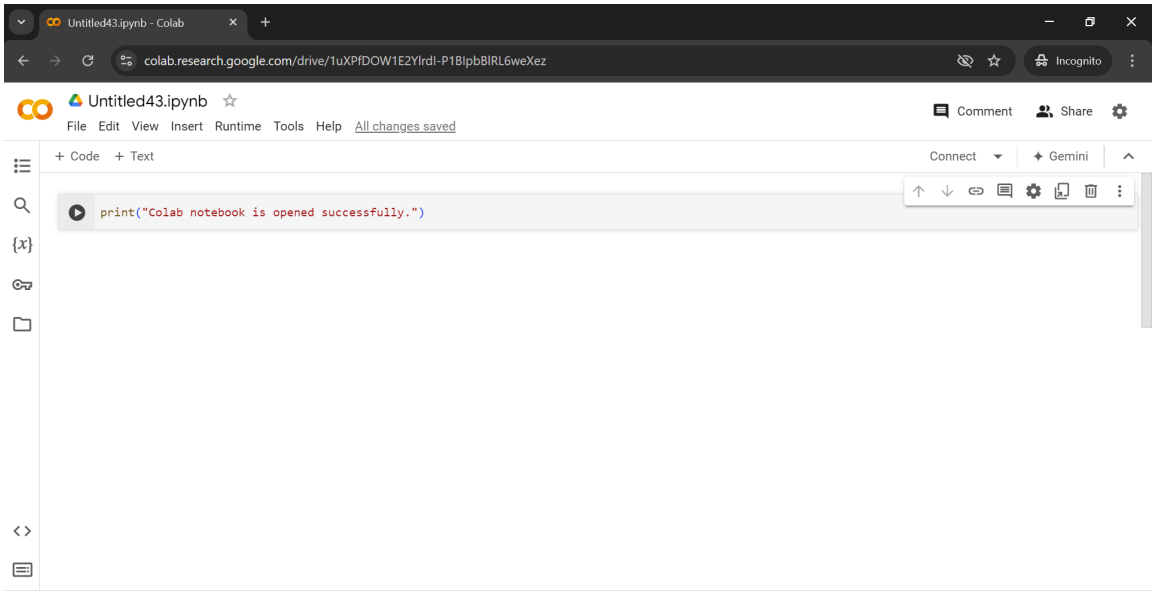
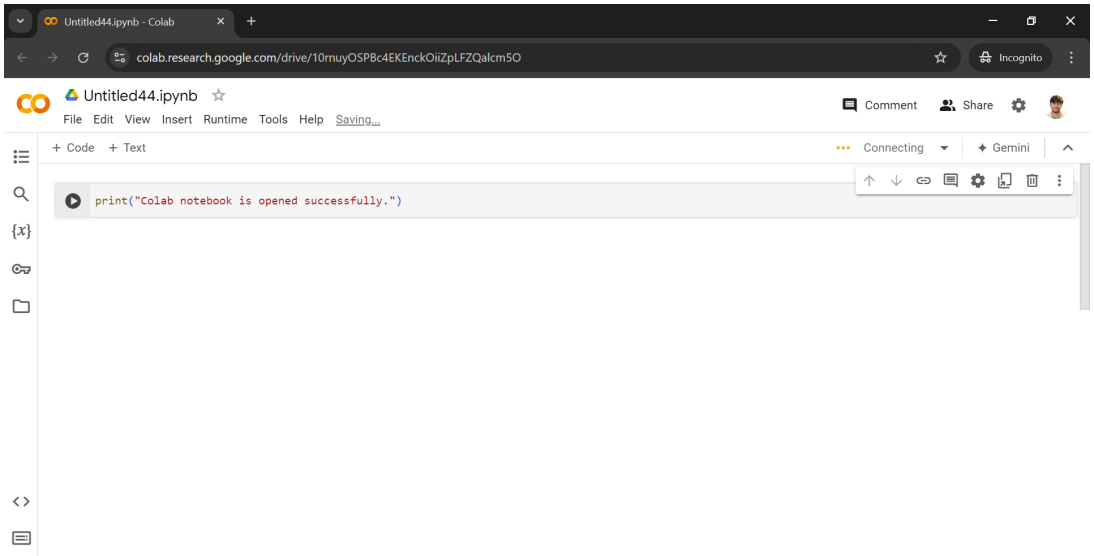
3. Type the following code for a test run.



Note: Google Colab offers a convenient server environment for executing programs. To utilize this, you need to establish an internet connection to access and run your code on Google Colab's servers.

4. Click on the **"Connect"** button at the top-right corner of the notebook interface. This will connect your notebook to a virtual machine or server provided by Google Colab that allows you to run your code.



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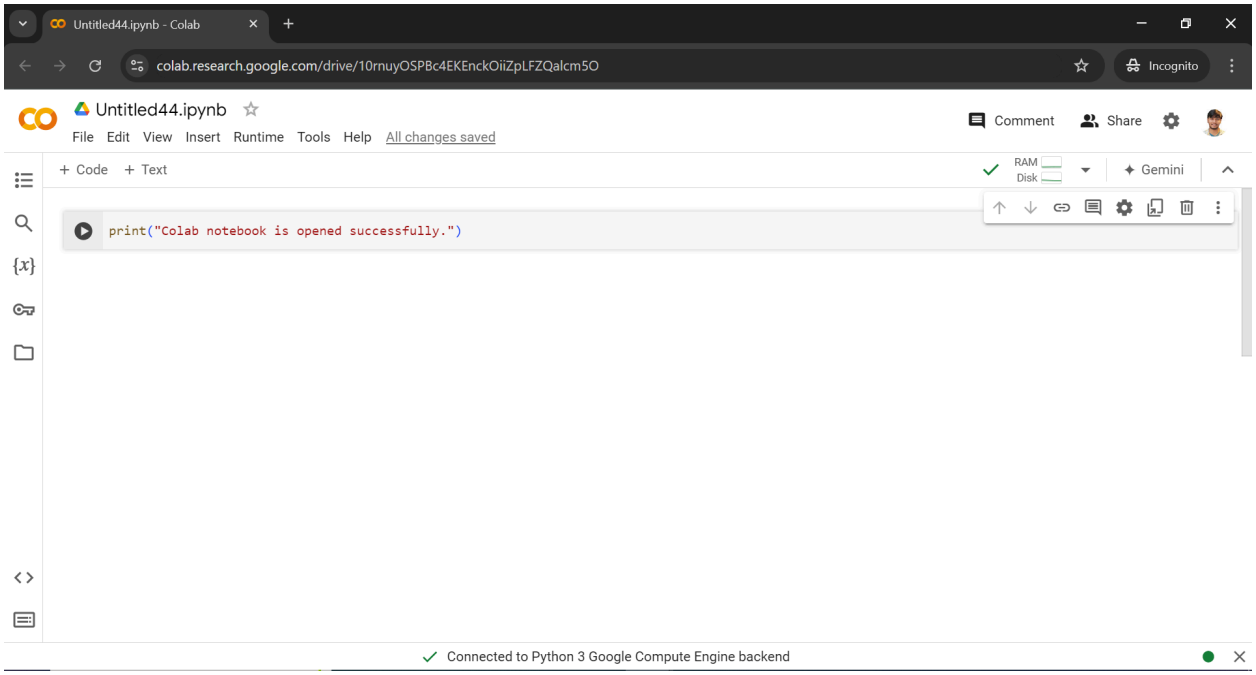
Note: After clicking the "Connect" button, the system will initiate a connection to the server. This process may take a few moments. While connecting, you'll notice three dots on the screen that gradually change color from orange to green, indicating the connection progress. A solid green color signifies a successful connection. Please note that connection times can vary based on network conditions and server load.

5. Once the connection is established, a green tick will appear, and the RAM and Disk usage indicators will become active.

Note: Additionally, a confirmation message stating "Connected to Python 3 Google Compute Engine backend" will be displayed at the bottom of the screen.



6. To run the code within a code cell, locate the run button (which has a 'Play'-type icon) positioned to the left of the cell and click it.

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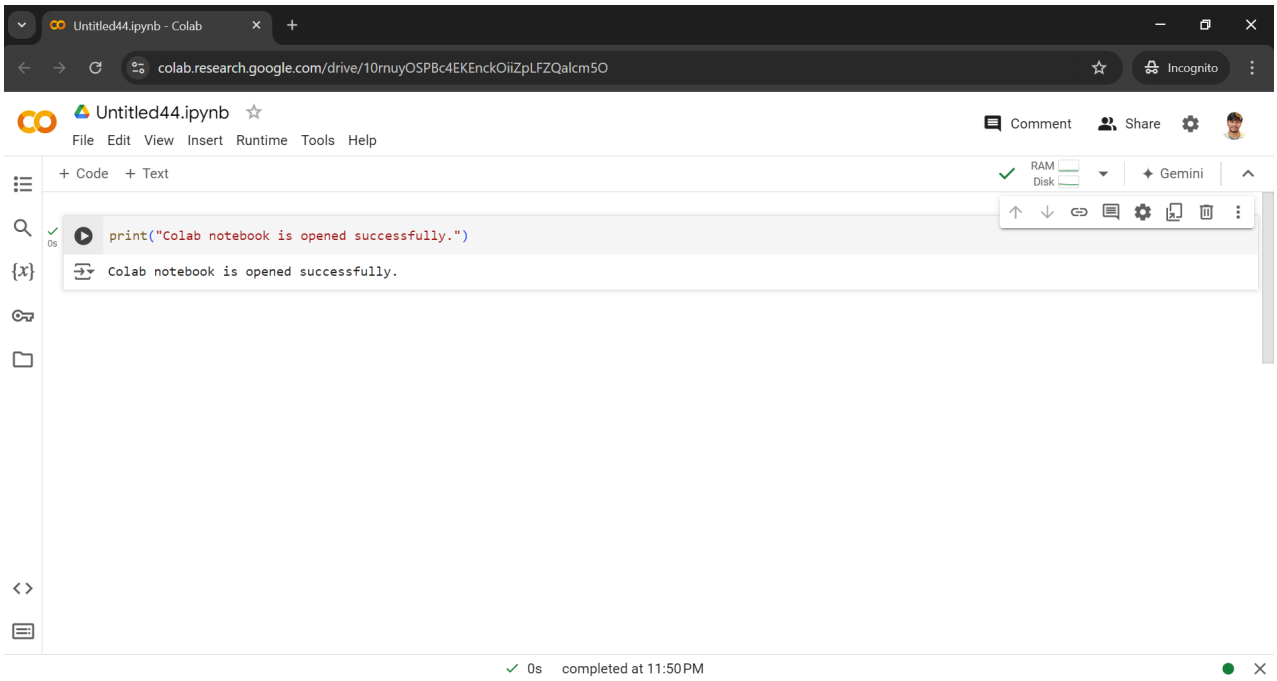

Upon successful code execution, a green tick will appear to the left of the code cell, and the resulting output will be displayed directly beneath it.

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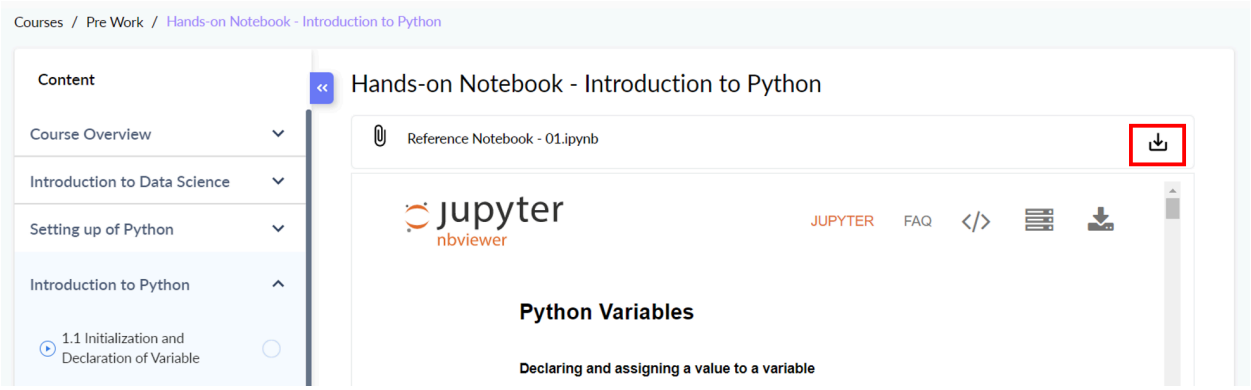
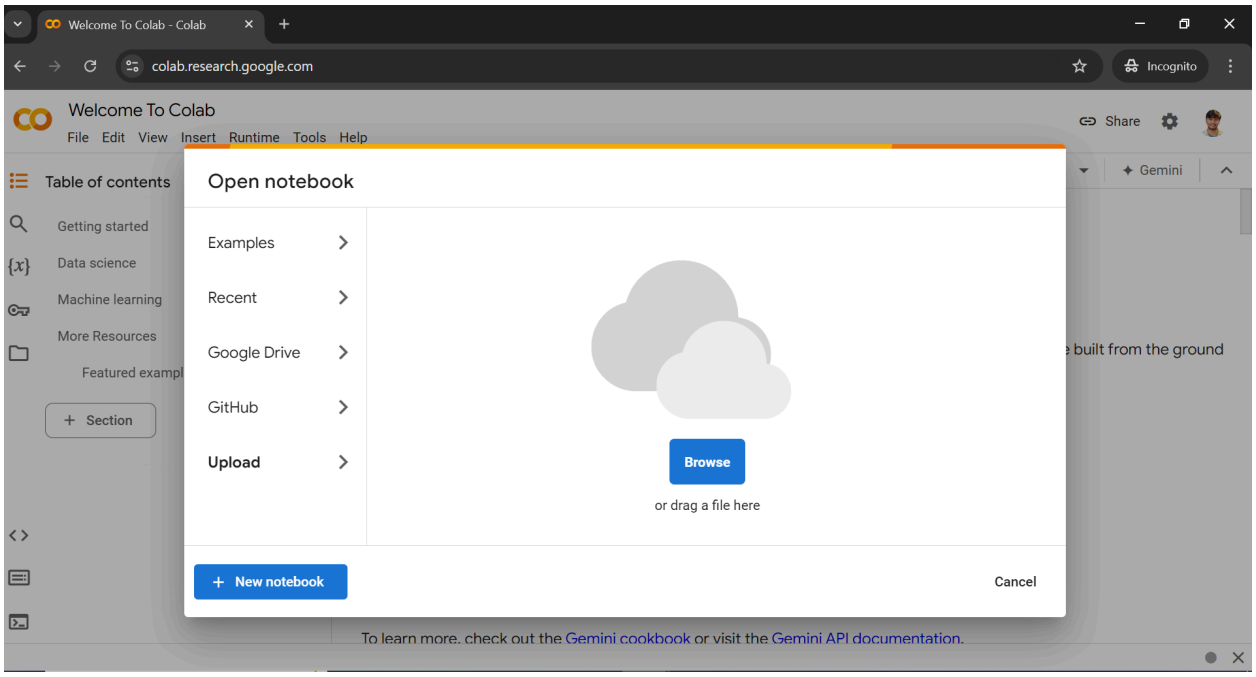
Uploading an existing Python notebook

1. Download a Python notebook (a file with a '.ipynb' extension) from Olympus. For example, click on the highlighted button in the image below to download the file.



Note: The above file, once downloaded to the local system, will NOT open in the desired manner by double-clicking it.

2. To upload the downloaded notebook, locate the **"Upload"** option in the left-hand panel of Google Colab and click on it.



3. Select **"Browse"** to navigate and choose the desired file from your local system.

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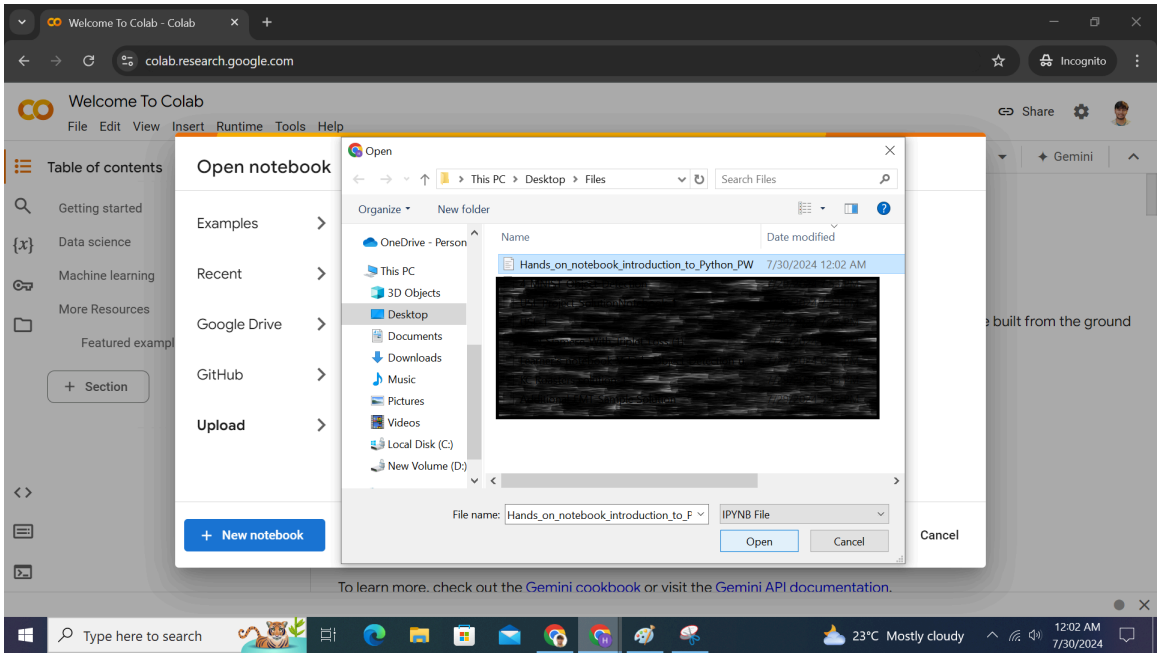
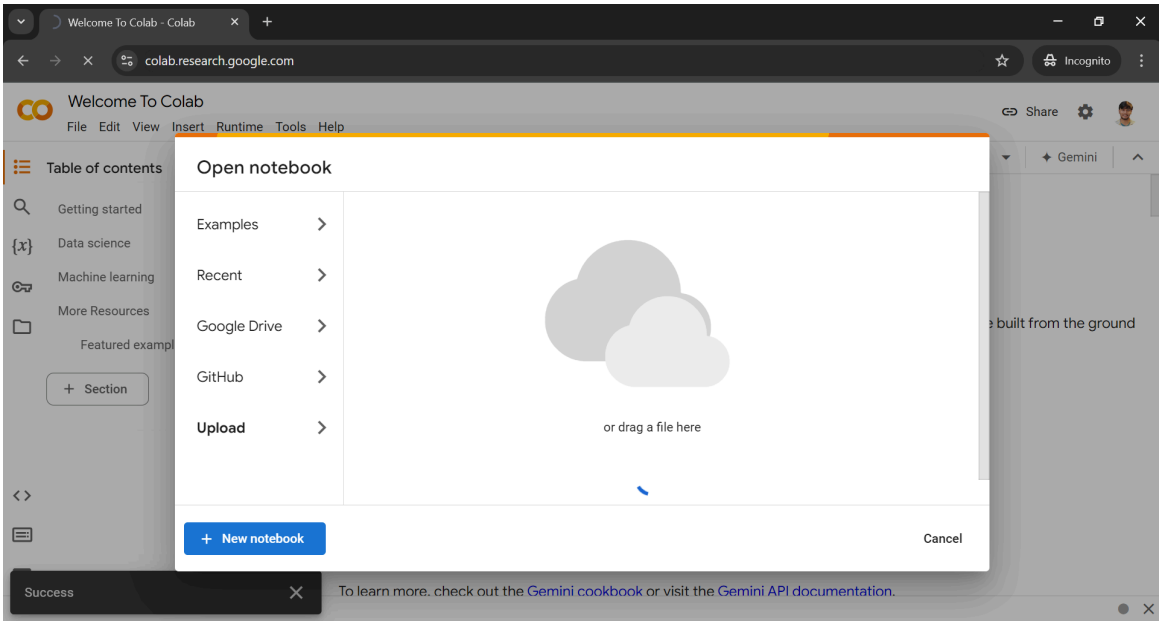
4. A standard file upload dialog box will open. Navigate to the location of your downloaded notebook, select the file, and then click the "**Open**" button.



Note: The notebook upload process can take a few moments to complete, depending on the file size. Once the upload is finished successfully, you will see a confirmation message or indicator, typically located in the bottom left corner of the screen.



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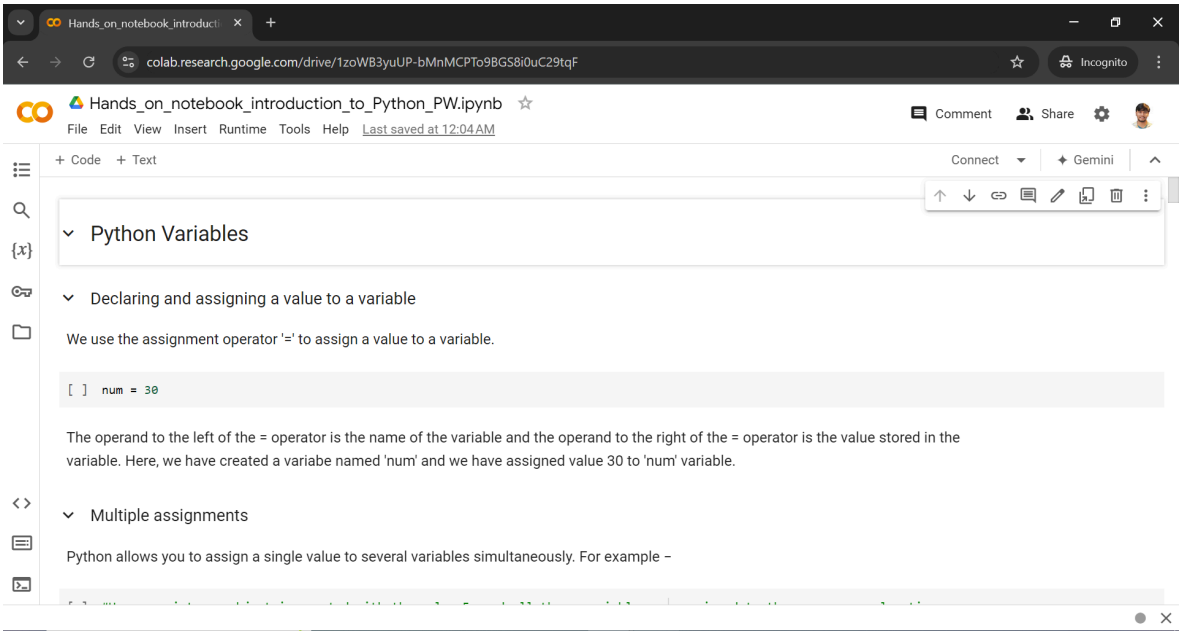
5. The uploaded notebook will now be visible in your Colab environment.



6. The notebook can now be edited or run as it is after connecting to a server.

Note: The steps to connect to a server are the same for an uploaded notebook as it is for one created in Colab.

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